0922 Ch.1 (Notes by Zi-Seok Lee)

$\boxed{0}$ History and Opening example

Poisson

B.C. — Games of chance in ancient Greece and Rome
17C — Fermat & Pascal
18C — Jacob Bernoulli derives the law of large numbers
De Moivre introduces the normal distribution and the CLT
19C — Laplace generalizes the CLT
Legendre and Gauss apply the least squares method
Poisson introduces the Poisson distribution
Chebyshev, Markov, Lyapunov study limit theorems
20C — Kolmogorov introduces the axiomatic system for prob. theory

1D Example Counting α particles $\alpha \to$: 2 protons + 2 neutrons

$N_t :=$ # of clicks during the time interval $[0, t)$

One way to derive $\mathbb{P}(N_t = r)$ is to divide $[0, t)$ into n subintervals

$$\mathbb{P}(N_t = r) = \mathbb{P}(r \text{ intervals with single clicks, } n-r \text{ without}) + \mathbb{P}(\text{Some intervals have 2 or more clicks})$$

Assumptions
- If we choose a large n, the second probability is negligable.
- Memoryless property (each subinterval is indep. with each other)
- $\mathbb{P}(\text{Click during } [a, b)) \propto (b-a)$ interval length
  $\mu(b-a)$

$$\Rightarrow \mathbb{P}(N_t = r) = \binom{n}{r}\left(\frac{\mu t}{n}\right)^r\left(1-\frac{\mu t}{n}\right)^{n-r} \xrightarrow[n\to\infty]{} \frac{(\mu t)^r}{r!}\cdot e^{-\mu t}$$

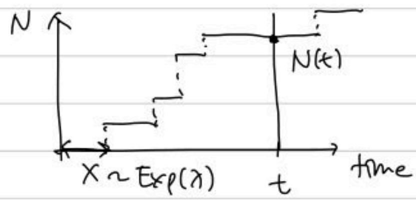
$$\therefore N_t \sim \mathcal{P}(\mu t) \text{ with rate } \mathbb{E}\left[\frac{N_t}{t}\right] = \mu$$

t.m.i.

Waiting time becomes memoryless
Let $N(t) \sim Poi(\lambda t)$: # of events in interval $[0, t]$
and X : time until the first event

$\Rightarrow X \sim Exp(\lambda)$

$\left(\because P(N(t) = 0) = e^{-\lambda t} = P(X \geq t) \Rightarrow \underline{F_X(x) = 1 - e^{-\lambda x}}\right)$



$\to$ The time from any point until the next event is $\sim exp(\lambda)$

Memoryless Property of $X \sim Exp(\lambda)$

$$P(X > s+t \mid X > s) = P(X > t) \to$$ $Y \sim Geom(p)$ is the only discrete r.v. with this prop.

1 Poisson Distribution $\mathcal{P}(\mu)$

- $X \sim \mathcal{B}(n,p)$: binomial distribution (n indep. trials with p)

$$\Rightarrow \mathbb{P}(X=r) = \binom{n}{r} p^r (1-p)^{n-r}$$

$\downarrow$ $n \to \infty$ while holding $\mu = np$ constant

$$\lim_{n\to\infty} \frac{n(n-1)\cdots(n-r+1)}{r!} \cdot \left(\frac{\mu}{n}\right)^r \left(1-\frac{\mu}{n}\right)^{n-r}$$

$$= \frac{\mu^r}{r!} e^{-\mu} =: \pi_r(\mu)$$

<u>Def</u> $X \sim \mathcal{P}(\mu) \Leftrightarrow \mathbb{P}(X=r) = \pi_r(\mu)$ $(r \geq 0)$ and $X \in \mathbb{Z}_{\geq 0}$

- $$\mathbb{E}(X) = \sum_{n=0}^{\infty} n \frac{\mu^n}{n!} e^{-\mu}$$

$$= \mu \sum_{n=0}^{\infty} \frac{\mu^n}{n!} e^{-\mu} = \mu$$

- Sometimes we extend $\mathcal{P}(\mu)$ to contain $\mathcal{P}(0)$, $\mathcal{P}(\infty)$

- Moments of X : $\mathbb{E}(X^2)$, $\mathbb{E}(X^3)$?

For $z \in \mathbb{C}$ with $|z| \leq 1$, $|z^X| \leq 1$

$$\mathbb{E}(z^X) = \sum_{n=0}^{\infty} z^n \cdot \frac{\mu^n}{n!} e^{-\mu} = e^{\mu(z-1)}$$

$\left.\frac{d}{dz}\right|_{z=1}$

$$\left.\begin{aligned} \mathbb{E}(X) &= \mu \\ \mathbb{E}(X(X-1)) &= \mu^2 \\ \mathbb{E}(X(X-1)(X-2)) &= \mu^3 \end{aligned}\right\} \to \begin{aligned} \mathbb{E}(X^2) &= \mu^2 + \mu \\ \mathbb{E}(X^3) &= \mu^3 + 3\mu^2 + \mu \end{aligned}$$

- $$\sum_{k=0}^{n} \pi_k(\mu) = 1 - \int_0^{\mu} \pi_n(\lambda)\, d\lambda \qquad (\to 1 \text{ as } n \to \infty)$$

$$\because \frac{d}{d\mu}\pi_k(\mu) = \frac{d}{d\mu}\left(\frac{\mu^k}{k!} e^{-\mu}\right) = \frac{\mu^{k-1}}{(k-1)!} e^{-\mu} - \frac{\mu^k}{k!} e^{-\mu}$$

$$= \pi_{k-1}(\mu) - \pi_k(\mu)$$

$$\frac{d}{d\mu} \sum_{k=0}^{n} \pi_k(\mu) = -\pi_n \quad (\pi_{-1}(\mu) := 0)$$ & integrate from $0 \sim \mu$

- (Additivity) $X \sim \mathcal{P}(\lambda)$, $Y \sim \mathcal{P}(\mu)$ & X, Y are independent

$\Rightarrow X+Y \sim \mathcal{P}(\lambda+\mu)$

pf $\quad \mathbb{P}(X=r \text{ and } Y=s) \overset{\text{independence}}{=} \mathbb{P}(X=r)\,\mathbb{P}(X=s) = \dfrac{\lambda^r}{r!} e^{-\lambda} \dfrac{\mu^s}{s!} e^{-\mu}$

So, $\mathbb{P}(X+Y=n) = \sum_{r+s=n} \dfrac{\lambda^r \mu^s}{r!\,s!} e^{-(\lambda+\mu)}$

$= \left(\sum_{r+s=n} \binom{n}{r} \lambda^r \mu^s \right) \cdot \dfrac{1}{n!} \cdot e^{-(\lambda+\mu)}$

$= \dfrac{(\lambda+\mu)^n}{n!} e^{-(\lambda+\mu)}$

- (Countable Additivity) $X_1, X_2, \cdots$: indep & $X_i \sim \mathcal{P}(\mu_i)$

If $\Sigma \mu_i$ converges, $S = \Sigma X_i \sim \mathcal{P}(\Sigma \mu_i)$

If $\Sigma \mu_i \to \infty$, $S \sim \mathcal{P}(\infty)$

pf $\quad S_n = \sum_{i=1}^n X_i \sim \mathcal{P}(\overbrace{\textstyle\sum_{i=1}^n \mu_i}^{\sigma_n})$ (by induction)

$\forall r, \ \mathbb{P}\{S_n \le r\} = \sum_{k=0}^r \pi_k(\sigma_n) = 1 - \int_0^{\sigma_n} \pi_r(\lambda)\, d\lambda$: decreases as $n \nearrow$

Also, $\{S_1 \le r\} \supseteq \{S_2 \le r\} \supseteq \cdots \supseteq \{S_n \ge r\}$

with $\lim_{n\to\infty} \bigcap_{k=1}^n \{S_k \le r\} = \{S \le r\}$

$\therefore \mathbb{P}\{S \le r\} = \lim_{n\to\infty} \mathbb{P}\{S_n \le r\} = \lim_{n\to\infty} \sum_{k=0}^r \pi_k(\sigma_n)$

If $\sigma_n \to \sigma$ as $n\to\infty$, $\mathbb{P}\{S \le r\} = \sum_{k=0}^r \pi_k(\sigma)$

If $\sigma_n \to \infty$ as $n \to \infty$, $\mathbb{P}\{S \le r\} = 0 \quad \forall r$

- (Obtaining binomial / multinomial distributions by conditioning)

$S = X_1 + X_2 + \cdots + X_n$ where $X_i \sim \mathcal{P}(\mu_i)$ & $X_1, \cdots, X_n$ are independent.

Then $S \sim \mathcal{P}(\sigma)$ where $\sigma = \mu_1 + \cdots + \mu_n$

$\Rightarrow \mathbb{P}(X_1 = r_1, \dots, X_n = r_n \mid S = s)$

$= \prod_{i=1}^n \dfrac{\mu_i^{r_i}}{r_i!} e^{-\mu_i} \Big/ \dfrac{(\sigma)^s}{s!} e^{-\sigma}$

$= \binom{s}{r_1, r_2, \cdots, r_n} \cdot \left(\dfrac{\mu_1}{\sigma}\right)^{r_1} \cdot \left(\dfrac{\mu_2}{\sigma}\right)^{r_2} \cdots \left(\dfrac{\mu_n}{\sigma}\right)^{r_n}$: multinomial distribution

For $n=2$,

$\mathbb{P}(X=r, Y=n-r \mid X+Y=n) = \binom{n}{r} p^r (1-p)^{r-1}$ where $p = \dfrac{\mu_X}{\mu_X + \mu_Y}$

$\sim \mathcal{B}(n, p)$

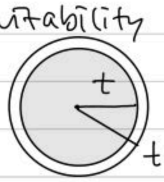
- (Obtaining Poisson variables from binomial conditional distribution)

$$\left.\begin{array}{l} X \sim \mathcal{P}(\mu) \\ Y|X \sim \mathcal{B}(X, p) \end{array}\right. \Rightarrow \ Y \text{ and } X-Y \text{ are independent Poisson variables}$$

pf
$$\begin{aligned} \mathbb{P}(Y=r,\ X=n) &= \mathbb{P}(Y=r \mid X=n)\ \mathbb{P}(X=n) \\ &= \binom{n}{r} p^r (1-p)^{n-r} \cdot \frac{\mu^n}{n!} e^{-\mu} \\ &= \underbrace{\frac{(\mu p)^r}{r!} e^{-\mu p}}_{\sim \mathcal{P}(\mu p)} \cdot \underbrace{\frac{(\mu(1-p))^{n-r}}{(n-r)!} e^{-\mu(1-p)}}_{\sim \mathcal{P}(\mu(1-p))} \end{aligned}$$

(Sum over all $n \geq r$)

$\boxed{2}$ The inevitability of the Poisson distribution

A_t : 
$\rightsquigarrow$
$$\left.\begin{array}{l} p_n(t) = \mathbb{P}\{N(A_t) = n\} \\ q_n(t) = \mathbb{P}\{N(A_t) \leq n\} \end{array}\right) \begin{array}{c} \text{differentiable} \\ \text{a.e.} \\ (\because \text{only jump} \\ \text{discontinuities}) \end{array}$$

Consider $N(A_t)$ when it jumps from n to $n+1$.

$\mu(t) = \mathbb{E}\{N(A_t)\} = \sum_{n=0}^{\infty} n \cdot p_n(t)$

$$\mathbb{P}\{\exists \text{point} \in A_{t+h} \setminus A_t \mid N(A_t) = n\} = \mu(t+h) - \mu(t)$$

↑ suppose this is <u>independent</u> of $N(A_t)$

$\mathbb{P}\{$jump occurs given $N(A_t)=n\}$: $q_n(t) - q_n(t+h) = p_n(t) \cdot \{\mu(t+h) - \mu(t)\}$

Letting $h \to 0$, $\quad -\frac{d q_n(t)}{dt} = p_n(t) \cdot \frac{d\mu(t)}{dt}$

From $q_0 = p_0$, $\ p_n = q_n - q_{n-1}$,

$$-\frac{dp_0}{dt} = p_0 \frac{d\mu}{dt} \quad \text{and} \quad -\frac{dp_n}{dt} = \frac{d}{dt}(q_n - q_{n-1}) = (p_n - p_{n-1})\frac{d\mu}{dt}$$

$\Updownarrow$ $\qquad\qquad$ $\Updownarrow \times e^{\mu}$

$$\frac{d}{dt}(\log p_0 + \mu) = 0 \qquad\qquad \frac{d}{dt}(p_n e^{\mu}) = p_{n-1} e^{\mu} \frac{d\mu}{dt}$$

$p_0(0) = 1,\ \mu(0) = 0 \Rightarrow \underline{p_0(t) = e^{-\mu(t)}}$ $\qquad$ $p_n(0) = 0 \quad (n \geq 1)$

$$\underline{p_n(t) = e^{-\mu(t)} \int_0^t p_{n-1}(s)\, e^{\mu(s)} \frac{d\mu}{ds}\, ds}$$

Inductively, $\quad p_1(t) = e^{-\mu(t)} \int_0^t e^{-\mu(s)} e^{\mu(s)} \frac{d\mu}{ds} ds = \mu(t)\, e^{-\mu(t)}$

$\vdots$

$$p_n(t) = \frac{\mu(t)^n}{n!} e^{-\mu(t)} = \mathbb{P}(N(A_t) = n)$$

$\therefore\ N(A_t) \sim \mathcal{P}(\mu(t))$

An inevitable consequence of the 'complete randomness' inherent in our assumptions of independence. (More in Ch.8)

$\boxed{3}$ Probability Spaces for Poisson Processes

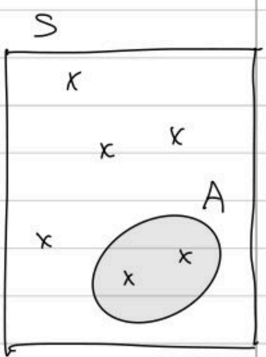
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

- Ω: "Sample space" of outcomes
- $\mathcal{F}$: "events" $\subset 2^{\Omega}$, σ-field on Ω
- $\mathbb{P}$: probability measure $\mathbb{P}: \mathcal{F} \to [0,1]$

Def A σ-field (or σ-algebra) on Ω is a nonempty collection $\mathcal{F}$ of subsets of Ω closed under complement, countable union, and countable intersections.
$(\Omega, \mathcal{F})$ is called a measurable space.

Def A real variable $X: \Omega \to \mathbb{R}$ is a measurable real function.
i.e., $\{X \le r\} \in \mathcal{F} \quad \forall r \in \mathbb{R}$

Def A random process is a collection of random variables $\{X_n\}_{n\in T}$
where each $X_n \in S$ and (S, Σ) is a measurable space.
(S: "State space"; finite or countable)
or $\Pi : \Omega \longrightarrow S^{\infty}$ (S^{∞} : countable subset of S)



If A is a 'test set' in S,

$$N(A) = \#\{\Pi(\omega) \cap A\} \quad : \Omega \longrightarrow \{0, 1, 2, \cdots, \infty\}$$

If $N(\cdot)$ is measurable for any A, $N(A)$ are random variables.
i.e., $\{\omega : N(A) = n\} \in \mathcal{F}$

ex $S = \mathbb{R}$, $A = (a.b)$ $\Rightarrow$ Any open set $G = \bigcup^{\infty} \underset{\text{disjoint}}{A_j}$, $N(G) = \sum^{\infty} N(A_j)$ is a r.v.
Any closed set $F = \bigcap^{\infty} G_i$ $(G_i \supset G_{i+1})$,
$\vdots$ $\quad N(F) = \lim_{i\to\infty} N(G_i)$
$\to$ For any Borel set A, $N(A)$ is a random variable

ex $S = \mathbb{R}^2$, $A = (a.b) \times (c,d)$ $\quad$ ($\cdots$ measure theory $\cdots$)

〈 To be continued 〉

Section 2.1. · Formal definition of a Poisson process $\Pi \in S^{\infty}$
where $N(A)$ are indep. ($N(A_1), N(A_2)$ are indep if $A_1 \cap A_2 = \emptyset$)
$N(A) \sim \mathcal{P}(\mu(A))$ where μ is a non-atomic measure on S

Section 2.5 · Conversely, for any non-atomic measure μ on S, is there Π s.t.
$N(A) \sim \mathcal{P}(\mu(A))$?
$\hookrightarrow$ Explicit construction is given with easily satisfied conditions (S, μ)

$\rightarrow$ A theory of great generality and wide applicability